**DTU Compute**
Department of Applied Mathematics and Computer Science

# Occupancy-Aware Climate Control Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis



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Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis
7 December, 2026

By
Mads Holme Håkonsson

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## Approval

This thesis has been prepared over six months at the Section for Indoor Climate, Department of Civil Engineering, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

Mads Holme Håkonsson - s234936

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*Signature*

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*Date*

## Abstract

ABSTRACT IS MISSING

## Acknowledgements

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# 1 Introduction

Reduction in the usage of energy in buildings is a crucial aspect if one wants to go towards a more sustainable future. It has long been known that buildings are responseible for X% of the total energy consumption in the world, and that a large part of this energy is for HVAC systems [MISSING REFERENCE] HVAC systems energy consumption has been directly related to the occupant behaviours of the buildings. This imposes a challenge, as one has to be able to understand the behaviour of the occupants in the buildings before we can gain the largest optimization in energy consumption. (MISSING REFERENCE) has shown that there is a large variability in the behavour of occupants such as thermal comfort, lighting & ventiliation needs and precense hours.

Within the context of energy consumption in buildings, heating has been estimated to contribute to X% [MISSING REFERENCE] of the consumption. An aspect of heating of a building is the byproduct of the thermal comfort of the occupants in the building, and therefore, if we can understand the thermal comforts of the occupants in the buildings, we can optimize the energy consumption drasitically. However, the thermal comfort curves is a personal preference of the occupants, and we should therefore expect this to differer dependent on the individual buildings and the number of occupants.

This projects aims to understand when the occupants of a single offices are present, and how much thermal comfort feedback does we need to collect from the occupants in order to be able to infer the true thermal comfort of the occupants in the building. The goal is to also build a codebase others and easyly download and extent.
We will do this by using a Hidden Markov Model (HMM) to model the state of the room

WRITTEN OUTLINE OF EACH CHAPTER AND WHAT QUESTIONS THEY ANSWER

# 2 Data

The data used in this project are $CO_2$ data collected from the office building 303B and the censors and data providers are climify [MISSING REFERENCE]. The data is collected in the fall and winter period of 2023, and the raw data period spans from [MISSING REFERENCE] to [MISSING REFERENCE]. The data collected from the censors are in intervals of around 10 minutes. Because our HMM models works in discrete time, the data have been aggregated into 30 minutes such that each time step $\Delta t = 1$ represents 30 minutes.

When collecting $CO_2$ data, the censors tend to drift over time [MISSING REFERENCE], and therefore, the data collected from the censors are not always accurate. To empty rooms, that have the same $CO_2$ levels may show different readins based on how much the censors have drifted over time. This is a problem, as we want to use the $CO_2$ data to infer the presence of the occupants in the building and a higher baseline of the $CO_2$ data may lead to a false positive of the presence of the occupants in the building. To correct for this, we have used a method called *Automatic Baseline Correction* (ABC) Algorithm. [MISSING REFERENCE]. In the use of this algorithm, we have assumed that the lowest $CO_2$ ppm values over a period is equal to the outside $CO_2$ ppm values. From literature, it's common to set the outside $CO_2$ ppm values to 400 ppm [MISSING REFERENCE], which we have also done.

Let $C_t$ be the $CO_2$ ppm value at time $t$, $\tilde{C}_t$ be the corrected $CO_2$ ppm value at time $t$.

We then define the modified ABC algorithm as

$$\tilde{C}_t = C_t - Q_{0.02}\left(C_{t-\Delta t:t}\right) + 400, \tag{2.1}$$

where $Q_{0.02}\left(C_{t-\Delta t:t}\right)$ is the 2% quantile of the $CO_2$ ppm values over the time period $[t - \Delta t, t]$ and $\Delta t$ is a fixed time period. In this project, we have set $\Delta t = 8$ days, such that the weekday overlaps. The reason for using a 2% quantile is to avoid the effect of erradic readings of the $CO_2$ ppm values, which may occur if the censors are not working properly. Because we are interrested in predicting future presence of the occupants in the building,

we dont need the data to be as granular as 10 minutes.

| Room | N | Min | 2.5% | Median | Mean | 97.5% | Max |
|---|---|---|---|---|---|---|---|
| Room 001 | 14289 | 341.00 | 426.00 | 621.00 | 828.52 | 2411.00 | 3185.00 |
| Room 003 | 14328 | 335.00 | 405.00 | 516.00 | 611.92 | 1424.00 | 3025.00 |
| Room 004 | 14268 | 292.00 | 400.00 | 940.50 | 1201.19 | 3042.00 | 3710.00 |
| Room 005 | 14324 | 345.00 | 418.00 | 704.00 | 774.84 | 1500.00 | 2447.00 |
| Room 007 | 14334 | 360.00 | 414.00 | 633.00 | 736.45 | 1588.00 | 2805.00 |
| Room 008 | 14321 | 360.00 | 429.00 | 781.00 | 988.87 | 2824.00 | 3581.00 |
| Room 009 | 14346 | 325.00 | 417.00 | 685.00 | 935.13 | 2322.50 | 3654.00 |
| Room 012 | 14344 | 335.00 | 414.00 | 739.00 | 769.75 | 1406.00 | 2860.00 |
| Room 013 | 14363 | 247.00 | 442.00 | 716.00 | 852.76 | 1841.00 | 2579.00 |
| Room 014 | 14144 | 327.00 | 422.00 | 804.00 | 1098.36 | 3305.42 | 4000.00 |
| Room 015 | 14336 | 337.00 | 414.00 | 754.50 | 985.40 | 2706.25 | 3808.00 |
| Room 016 | 14333 | 349.00 | 422.30 | 686.00 | 833.00 | 1841.70 | 3825.00 |
| Room 019 | 14338 | 355.00 | 435.00 | 727.00 | 886.71 | 1927.57 | 2756.00 |

Table 2.1: Summary statistics of the raw $CO_2$ readings (ppm) per room, after removing missing, saturated and duplicate readings.

TO DO:

- Make summary table of the data with before and after the ABC algorithm (n, mean, sd, min, quantiles, max, % of bins below 450 ppm; one row per room if several rooms).
- Make ACF plots to show the AR structure of the data. Add PACF (shows AR order); note daily cycle at lag 48.
- Make a plot of the data before and after the ABC algorithm (raw, estimated floor, calibrated + zoom on one week).
- Histogram/density of calibrated $CO_2$ (multimodality motivates hidden states).
- Average daily profile, weekday vs. off-day (motivates off-day + Fourier covariates).
- Overview plot of full series with train/val/test boundaries and feedback events.
- Short chapter introduction and conclusion.

Fix in existing text:

- $\Delta t$ used both as 30-min step and as 8-day window: rename window to $W$.
- Dates: Room 009 segment is 2023-11-21 to 2024-02-09 (3860 bins, $\approx$ 80 days), not "fall and winter 2023". Clarify 303B vs. DTU dataset (live `room_009/30min` data comes from `clean_dtu_data.py`).
- Reason for 8 days: every window contains a full weekend (an empty-room period).
- Merge the duplicated aggregation paragraph; state that bins are means of the readings.
- Typos: censors $\rightarrow$ sensors, "To empty rooms" $\rightarrow$ Two, readins, erradic, interrested, dont, climify $\rightarrow$ Climify.
- Define a `\COtwo` macro instead of retyping `$\text{CO}_2$`.

Missing processing steps to describe:

- Raw cleaning: drop NaNs, keep 0–4000 ppm (sensor saturation), remove duplicates.
- Gap handling: only the longest gap-free segment is kept (discrete model needs equal spacing).
- ABC details: applied after 30-min binning (and why); trailing window, so no leakage into val/test; first 8 days dropped as warm-up; justify 2% quantile (must be below share of empty bins, $\approx$16% below 500 ppm in Room 009).
- Report floor drift (Room 009: 388 $\rightarrow$ 500 ppm).
- State ABC assumption: room is empty and ventilated to outdoor level at some point in every window. Outdoor $CO_2$ was $\approx$420 ppm in 2023–24: justify 400 or mention it.
- Feedback data: how collected, number of responses, binary flag per 30-min bin.
- Humidity: calibration against outdoor humidity, or state that it is excluded.

- Covariates: off-day flag (weekends + holidays), daily (3 harmonics) and weekly (1 harmonic) Fourier terms, weather variables; z-scored with training statistics only.
- Time zone: timestamps in UTC (no DST change in Nov–Feb).
- Split: chronological 65/15/20, boundaries 2024-01-12 and 2024-01-24; Christmas break falls in training.
- Rooms and sensors: which rooms and why, single/shared offices, ventilation type, sensor type (NDIR), accuracy, sampling interval.

Missing references: Climify, NDIR sensor drift, ABC (Senseair firmware; check whether your sensors already run it), outdoor $CO_2$ baseline (e.g. NOAA).

# 3 Theory

In this chapter, a ODE of the physical model of the $CO_2$ dynamics will be presented and how the process is descibed as a stochastic differential equation (SDE), where noise has been added to the system.
Given the presented SDE, it will be shown that the process can be desbribed as a discrete-time AR(1) process, where the production rate of the occupants in the room is modelled as a function of the occupancy state of the room.
Hereafter, the generel theory of the hidden markov models (HMM) will be presented along with the inference algorithms used in this project.

## 3.1 Modelling $CO_2$ dynamics

This subsection present the underlying physical model of the $CO_2$ dynamics of a room and show that it can be described as a discrete-time AR(1) process and how the production rate of the occupants in the room can be modelled as a function of the occupancy state of the room.

### 3.1.1 The underlying physical model & assumptions

We adopt the mass balance model from litterature (Batterman 2017, Wang, Burnett, and Chong 1999), the mass balance equation of $CO_2$ in a room can be described as a ordinary differential equation (ODE) of the form

$$V\frac{dC_{\text{indoor}}(t)}{dt} = P(t) + \alpha V\left(C_{\text{outdoor}}(t) - C_{\text{indoor}}(t)\right), \tag{3.1}$$

where $V$ is the volume of the room ($m^3$), $C_{\text{indoor}}(t)$ is the indoor $CO_2$ concentration at time $t$ ($mg{\cdot}m^{-3}$), $C_{\text{outdoor}}(t)$ is the outdoor $CO_2$ concentration at time $t$ ($mg{\cdot}m^{-3}$), $P(t)$ is the $CO_2$ production rate of the occupants in the room ($mg{\cdot}h^{-1}$), and $\alpha$ is the air change rate of the room ($h^{-1}$), such that $\alpha V$ is the outdoor air flow rate ($m^3{\cdot}h^{-1}$).

Since the sensors report $CO_2$ in ppm, measurements are converted using $C\,[\text{mg}\cdot\text{m}^{-3}] = C\,[\text{ppm}]\cdot M_{CO_2}/V_m$, where $M_{CO_2} = 44.01$ $g{\cdot}mol^{-1}$ and $V_m$ is the molar volume of air ($\approx 24.45$ $L{\cdot}mol^{-1}$ at 25°C and 1 atm).

The underlying physical model of the $CO_2$ dynamics of a room can be descibed as a odinary differential equation (ODE) of the form

$$\frac{d\tilde{C}(t)}{dt} = P(t) - \alpha\tilde{C}(t), \tag{3.2}$$

where $\tilde{C}(t) = C_{\text{indoor}}(t) - C_{\text{outdoor}}(t)$ is the $CO_2$ ppm value at time $t$, $P(t)$ is the $CO_2$ production rate of the occupants in the room, and $\alpha$ is the decay rate of the $CO_2$ in the room.
The decay rate $\alpha$ can also be interpreted as the air change rate of the room i.e. how much of the $CO_2$ in the room is ventilated out of the room at time $t$.

The assumptions in 3.2 is the following:

- The $CO_2$ in the room is well mixed, meaning that the $CO_2$ is evenly distributed in the room.
- There is no $CO_2$ sinks in the room, meaning that the $CO_2$ is not absorbed by any materials in the room and is only ventilated out of the room.

### 3.1.2 Diffusion Process (Ornstein-Uhlenbeck)

The underlying physical model of the $CO_2$ dynamics of a room can be described by a diffusion process. The diffusion process is a continuous-time stochastic process that describes the change of $CO_2$ in a room.
Let $\tilde{C(t)} = C(t) - \min_{\forall t}(C(t))$ such that the process reverts back to a baseline of 0. The diffusion process can then be desbribed as

$$\Delta\tilde{C(t)} = (P(t) - \alpha\tilde{C(t)})\Delta t + \sigma\Delta W(t), \tag{3.3}$$

where $P(t)$ is the $CO_2$ production rate of the occupants in the room, $\alpha$ is the decay rate of the $CO_2$ in the room, $\sigma$ is the diffusion coefficient and $W(t)$ is a Wiener process, which is interpretated as the noise in the sensors.
This models tells us, that if the production rate of the occupants at time $t$ is higher than the decay rate (or air change rate) of the room, then the $CO_2$ levels in the room will increase. The argument for this physical model that $CO_2$ accumate and that it does not decay by itself [MISSING REFERENCE], unless it's ventilated out of the room.

### 3.1.3 Proof of AR(1) process

We now want to show, that the underlying physical model of the $CO_2$ dynamics of a room can be described as a discrete-time AR(1) process.
Firstly, we recognize the diffusion process as a Ornstein-Uhlenbeck, and the solution of this process is given by Iacus (2008, Sec. 1.13.1) as

$$\tilde{C}(t) = \frac{P}{\alpha} + \left(x_0 - \frac{P}{\alpha}\right)e^{-\alpha t} + \theta_3 \int_0^t e^{-\alpha(t-u)}\,\mathrm{d}W_u \tag{3.4a}$$

or, in the second parametrization,

$$\tilde{C}(t) = \mu + (x_0 - \mu)e^{-\alpha t} + \sigma\int_0^t e^{-\alpha(t-u)}\,\mathrm{d}W_u, \qquad \mu = \frac{P}{\alpha}, \tag{3.4b}$$

where $x_0$ is the initial value of the process at time $t = 0$.

If we then discretize the process with a time step $\Delta t$, we can write the process as

$$\tilde{C}_{t+\Delta t} = \mu + (\tilde{C}_0 - \mu)e^{-\alpha(t+\Delta t)} + \sigma\int_0^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u, \tag{3.5a}$$

where we can split the integral into two parts as

$$\begin{aligned}\sigma\int_0^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u = \sigma\Bigg(&e^{-\alpha\Delta t}\int_0^t e^{-\alpha(t-u)}\,\mathrm{d}W_u \\ &+ \int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u\Bigg).\end{aligned} \tag{3.5b}$$

We can then write out the process as

$$\tilde{C}_{t+\Delta t} = \mu(1 - e^{-\alpha\Delta t}) + \mu e^{-\alpha\Delta t} + (\tilde{C}_0 - \mu)e^{-\alpha t}e^{-\alpha\Delta t}$$
$$+\sigma e^{-\alpha\Delta t}\int_0^t e^{-\alpha(t-u)}\,\mathrm{d}W_u + \sigma\int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u \tag{3.5c}$$

We see then that

$$\begin{aligned}\tilde{C}_{t+\Delta t} &= \left(\mu + (\tilde{C}_0 - \mu)e^{-\alpha t} + \sigma\int_0^t e^{-\alpha(t-u)}\,\mathrm{d}W_u\right)e^{-\alpha\Delta t} \\ &\quad + \mu(1 - e^{-\alpha\Delta t}) + \sigma\int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u \\ &= \tilde{C}_t e^{-\alpha\Delta t} + \mu(1 - e^{-\alpha\Delta t}) + \sigma\int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u \end{aligned} \tag{3.5d}$$

We have now shown that diffusion process is dependent on the previous emission $\tilde{C}_t$.
But for it to be an AR(1) process, we nedd to show that the drift term has mean 0, constant variance and is unoccorelated over time. [MISSING REFERENCE].
Let $\Delta t = c \in \mathbb{N}$ be a constant time step and given that $\alpha, t \in \mathbb{R}_{\geq 0}$, then from 3.5d, we can see that the drift term is given by

$$\sigma\int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u = \tag{3.6}$$

We recognize $e^{-\alpha(t+\Delta t-u)}$ as our stochastic term, which is determinitics, meaning that that its ito intergratable as $\int_0^T e^{-2\alpha(t+\Delta t-u)}\,\mathrm{d}W_u < +\infty$.
Therefore, from Iacus (2008, Thm. 1.9.1), We now know that the mean of the drit term is

$$\mathbb{E}\left[\sigma\int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u\right] = 0 \tag{3.7}$$

And that the variance is

$$\mathrm{Var}\left[\sigma\int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u\right] = \sigma^2\int_t^{t+\Delta t} e^{-2\alpha(t+\Delta t-u)}\,\mathrm{d}u \tag{3.8}$$

$$\mathrm{Var}\left[\sigma\int_t^{t+\Delta t} e^{-\alpha(t+\Delta t-u)}\,\mathrm{d}W_u\right] = \sigma^2\int_t^{t+\Delta t} e^{-2\alpha(t+\Delta t-u)}\,\mathrm{d}u \tag{3.9a}$$
$$= \sigma^2 e^{-2\alpha(t+\Delta t)}\int_t^{t+\Delta t} e^{2\alpha u}\,\mathrm{d}u \tag{3.9b}$$
$$= \sigma^2 e^{-2\alpha(t+\Delta t)}\left[\frac{1}{2\alpha}e^{2\alpha u}\right]_t^{t+\Delta t} \tag{3.9c}$$
$$= \frac{\sigma^2}{2\alpha}e^{-2\alpha(t+\Delta t)}\left(e^{2\alpha(t+\Delta t)} - e^{2\alpha t}\right) \tag{3.9d}$$
$$= \frac{\sigma^2}{2\alpha}\left(1 - e^{-2\alpha\Delta t}\right) \tag{3.9e}$$

We see that the variance is independent of $t$ and that its constant over time.
We now only need to show that the covariance of the drift is 0.
We define that time steps as $t_1, t_2 \in \mathbb{R}_{\geq 0}$ such that $t_1 < t_2$ and that the time steps are

disjoint, meaning that $[t_1, t_1 + \Delta t] \cap [t_2, t_2 + \Delta t] = \emptyset$ and that $t_1 + \Delta t = t_2$. We then define $e_k$ as the drift term at time $t_k$ such that

$$\begin{aligned} \mathbb{E}[e_k] &= 0 \\ \mathsf{Var}(e_k) &= \sigma \int_{t_{k-1}}^{t_k} e^{-\alpha(t+\Delta t-u)}\, \mathrm{d}W_u = \frac{\sigma^2}{2\alpha}\left(1 - e^{-2\alpha\Delta t}\right) \end{aligned} \tag{3.10}$$

We now then see that the covariance of the drift term is given by, given that $j \neq k$

$$\begin{aligned} \mathsf{Var}(e_k + e_j) &= \mathsf{Var}(e_k) + \mathsf{Var}(e_j) + 2\mathsf{Cov}(e_k, e_j) \\ &\Leftrightarrow \end{aligned} \tag{3.11}$$

To compute $\mathsf{Var}(e_k + e_j)$, we write both drift terms as integrals over the same interval $[0, T]$, where $T \geq \max(t_k, t_j) + \Delta t$, by introducing the deterministic functions

$$f_k(u) = \mathbb{1}_{[t_k,\, t_k+\Delta t)}(u)\, e^{-\alpha(t_k+\Delta t-u)}, \tag{3.12}$$

such that

$$e_k = \sigma \int_0^T f_k(u)\, \mathrm{d}W_u. \tag{3.13}$$

Since $j \neq k$, the intervals $[t_k, t_k + \Delta t)$ and $[t_j, t_j + \Delta t)$ are disjoint, and therefore

$$f_k(u) f_j(u) = 0 \quad \forall u \in [0, T]. \tag{3.14}$$

The sum $e_k + e_j = \sigma \int_0^T (f_k(u) + f_j(u))\, \mathrm{d}W_u$ is again an Itô integral with a deterministic integrand, so by Iacus (2008, Thm. 1.9.1)

$$\begin{aligned} \mathsf{Var}(e_k + e_j) &= \sigma^2 \int_0^T (f_k(u) + f_j(u))^2\, \mathrm{d}u \\ &= \sigma^2 \int_0^T f_k(u)^2\, \mathrm{d}u + \sigma^2 \int_0^T f_j(u)^2\, \mathrm{d}u + 2\sigma^2 \int_0^T f_k(u) f_j(u)\, \mathrm{d}u \\ &= \mathsf{Var}(e_k) + \mathsf{Var}(e_j) + 0. \end{aligned} \tag{3.15}$$

Comparing with $\mathsf{Var}(e_k + e_j) = \mathsf{Var}(e_k) + \mathsf{Var}(e_j) + 2\,\mathsf{Cov}(e_k, e_j)$ gives

$$\mathsf{Cov}(e_k, e_j) = 0, \qquad j \neq k. \tag{3.16}$$

Hence the drift terms have mean zero, constant variance and are uncorrelated, and the discretized process is an AR(1) process.

### 3.1.4 Modelling Production Rate

By assuming that the prodction rate of the occupants is only depented on the amount of occupants in the room, we can model the production rate as a function of the occupancy state of the room. Let $S(t)$ be the occupancy state of the room at time $t$, then we can model the production rate as

$$P(t) = \beta_{S(t)}, \tag{3.17}$$

where $\beta_{S(t)}$ is the production rate of the occupants in the room, which is dependent on the occupancy state of the room. This gives a stepwise prodiuction rate function, which is only dependnt on the occupancy state of the room.

## 3.2 Hidden Markov Model (HMM) theory

This section will present the underlying models and the theory behind it that models the diffusion process. Here to, inference algorithms and analysis methods will be presented.

### 3.2.1 The underlying theory

Hidden markov models are markov chains where the underlying states are hidden and only a random variable that is dependent on the underlying state is observed.
This can be described as a markov chain with a set of hidden states $S = \{s_1, s_2, ..., s_N\}$ and a set of observations $O = \{o_1, o_2, ..., o_M\}$. The transition between the hidden states is governed by a transition matrix $\Gamma$ and the observations are generated by an emission matrix $B$. The initial state distribution is given by $\pi$. The HMM stucture can then debsrib the emissions as

$$P(O_t|S_t) = B_{S_t,O_t}, \tag{3.18}$$

where $B_{S_t,O_t}$ is the probability of observing $O_t$ given the hidden state $S_t$. The transition between the hidden states is given by

$$P(S_t|S_{t-1}) = \Gamma_{S_{t-1},S_t}, \tag{3.19}$$

where $\Gamma_{S_{t-1},S_t}$ is the probability of transitioning from hidden state $S_{t-1}$ to hidden state $S_t$. The initial state distribution is given by

$$P(S_1) = \pi_{S_1}, \tag{3.20}$$

where $\pi_{S_1}$ is the probability of starting in hidden state $S_1$.

### 3.2.2 Likelihood

Several methods can be used to estimate the parameters of a HMM, where at some of the more common methods are the Baum-Welch algorithm [MISSING REFERENCE].
In this project, we will however use the method of maximum likelihood estimation (MLE) to estimate the parameters of the HMM with the forward-filtering algorithm [MISSING REFERENCE].

The forward algorithm implemented is called recursive filtering (Møller et al. 2024), as it constructs the current state distribution $p(X_t \mid Y_{1:t})$ recursively from the previous state distribution $p(X_t \mid Y_{1:t-1})$.

**Algorithm 1** Recursive Filtering Forward Algorithm

**Require:** Observations $\mathbf{y} = (y_1, \ldots, y_T)$, covariates $\mathbf{x} = (x_1, \ldots, x_T)$, initial state distribution $u_0 \in \mathbb{R}^{1\times N}$, HMM parameters $\theta$
**Ensure:** Predicted distributions $\{u_t\}$, filtered distributions $\{u_{t|t}\}$, step likelihoods $\{f_t\}$

**for** $t = 1, \ldots, T$ **do**
  $\Gamma_t \leftarrow \text{TransitionMatrix}(\theta, t, y_t, x_t)$ ▷ $N \times N$ transition matrix
  $u_{t|t-1} \leftarrow u_{t-1|t-1} \cdot \Gamma_t$ ▷ Predicted state distribution, $1 \times N$
  $g_t \leftarrow \text{Density}(\theta, t, y_t, x_t)$ ▷ Emission densities for all states, $1 \times N$
  $f_t \leftarrow \sum_{i=1}^{N} [u_t \odot g_t]_i$ ▷ Marginal likelihood at time $t$
  $f_t \leftarrow \max(f_t, 10^{-10})$ ▷ Clip to avoid division by zero
  $u_{t|t} \leftarrow \dfrac{u_t \odot g_t}{f_t}$ ▷ Filtered state distribution (normalized)
**end for**
**return** $\{u_t\}, \{u_{t|t}\}, \{f_t\}$

where $\odot$ denotes element-wise multiplication of vectors. The log-likelihood is obtained by summing the log of the step likelihoods:

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} \log f_t. \tag{3.21}$$

Each likelihood step $f_t$ is the sum of the predicted state distribution $u_t$ multiplied by the emission densities $g_t$ for all states, which gives the marginal likelihood of observing $y_t$ given the previous observations $y_{1:t-1}$.

### 3.2.3 Forward Backward Filtering
### 3.2.4 Backward Sampling
### 3.2.5 Model Diagnostics

# 4 Implementation

# 5 Model Results

# 6 Simulation Study

A

# 7 Discussion

# 8 Conclusion

# 9 Future Work

# Bibliography

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# A Title

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